

Maged Hennawy

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LEAD FULL-STACK & PLATFORM ENGINEER

Senior full-stack engineer with 8 years of experience architecting and delivering enterprise-scale applications. Specialized in modern SPA architecture (React, Vue, Angular) and backend API development (Node.js, PostgreSQL), owning the full lifecycle from secure cloud infrastructure and CI/CD to heavily-typed user interfaces. Most recently the owner of a shared platform behind ~10 secure government applications, with production LLM/RAG features on AWS Bedrock. Founder of a live fintech product.

Experience

JAN 2025 – PRESENT

Lead Full-Stack & Platform Engineer / Royal Canadian Air Force, Ottawa (remote)

- Architected and own RAP, a start-to-production platform that cut new-project setup and deployment from weeks to under an hour.
- Terraform and AWS CDK for ECS Fargate, Aurora, Cognito, KMS and Secrets Manager; led a CDK → Terraform consolidation.
- SSO via Microsoft Entra ID (Cognito federation) with Casbin RBAC/ABAC; least-privilege IAM and automated secret rotation.
- Model Context Protocol server with 10+ tools orchestrating embeddings, vector search and LLM inference.
- RAG over pgvector with source attribution, cited sources and confidence scores, streamed over SSE.
- Replaced a legacy Excel/VBA solver with an $O(n \log n)$ constrained-optimization engine that scales past 1,000+ activities and guarantees coverage of all 25 funding segments.
- Continuous ATO compliance platform (NIST 800-53, 59 custom cloud checks) and a 770-finding vulnerability remediation across 400+ resources.
- Technical lead for an 8-person pod; hiring for subcontractors and mentoring engineering interns.

Technologies: Vue 3 / React / TypeScript / Node.js / PostgreSQL / Terraform / AWS / AWS Bedrock / MCP / pgvector / Casbin

DEC 2021 – JAN 2024

Full Stack Engineer / Clarify Health, San Francisco (remote)

- Scalable Node.js API endpoints handling 10,000+ requests per minute at 99.99% uptime.
- Extensible data components inside a plug-and-play framework configured by customer-success teams.
- Documentation for the UI library and API services that cut developer onboarding time by 40%.

Technologies: Angular / RxJS / Node.js / PostgreSQL / AWS Athena / Redshift / Redis

JAN 2020 – DEC 2021

Senior Angular Developer, Lead Developer / RBC, Toronto, ON

- Built a UI/UX library for Vue, Angular and React that RBC web products standardized on.
- Accessibility and localization work (WCAG) that grew the library's adoption by 30%.
- Led integration of third-party APIs and microservices with the Node.js backend.

Technologies: Angular / React / Vue / NgRx / ASP.NET Core / MS SQL Server

MAY 2017 – JAN 2020

Software Developer / IBM, Markham, ON

- Started IBM Carbon Components Angular (28,000+ downloads). Still a code owner and reviewer.
- Built the UI/UX component library for the Watson business unit with designers and developers.
- Implemented globalization and localization for international users.

Technologies: Angular / TypeScript / NgRx / ASP.NET Core / Storybook / Jest

Education

Honours BSc, Software Engineering / University of Toronto

Advanced ReactJS / Meta (Coursera)

Ventures & open source

- **Perus (Founder)** (perus.ca): Live Canadian real-estate equity-crowdfunding platform. Three tiers (Astro site, Vue app, Hono API) on Cloudflare Workers + Fly.io, with securities-exemption-aware architecture, Canadian data residency and a points-and-rewards engine.
- **ZeroTax (Builder)**: Multi-entity Canadian tax-optimization engine for CCPCs and a personal T1: CRA T5 XML e-file, SR&ED flagging, dividend/SBD modelling and Gemini-powered transaction categorization.
- **Jinx (Builder)**: Employee payroll and scheduling app for a small business, with real-time hours tracking and automatic payroll calculation.
- **Carbon Components Angular (Active code owner / reviewer)**: Maintaining and reviewing code for the open-source IBM UI library I started (28,000+ downloads).

Skills

Front-end: TypeScript, JavaScript, Angular, React, Vue, Next.js, Vuetify, Tailwind CSS, Highcharts, Mapbox, HTML5, SCSS

Back-end / data: Node.js, Express, FastAPI (Python), Loopback, Hono, .NET Core (C#), Sequelize, Prisma, PostgreSQL, MS SQL Server, MongoDB

Cloud & DevOps: AWS (ECS, RDS/Aurora, Cognito, KMS, S3, Athena/Redshift), Microsoft Entra ID, Terraform, AWS CDK, Cloudflare, Fly.io, Docker, cosign, Semgrep / Trivy / SonarQube, Casbin

AI / ML: AWS Bedrock (Claude), Model Context Protocol (MCP), RAG with citations, embeddings & vector search (pgvector), SSE / WebSocket streaming, constrained optimization

Workflow: Git, GitHub, Azure DevOps, CircleCI, Jest, Jasmine, Karma, Mocha, Chai, Storybook